

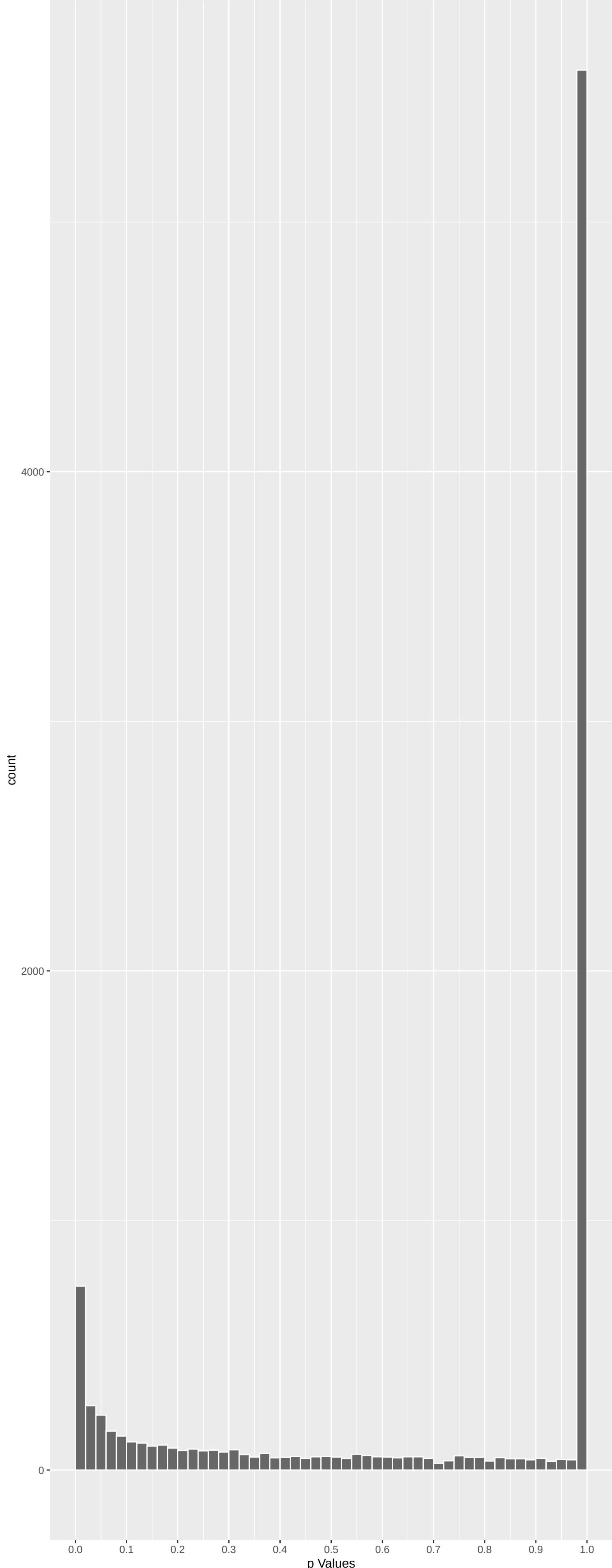
Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
IQGAP3	-6.491090	5.113050e-10	7.723e-06	1.166e-01
PROM1	5.975173	1.379080e-08	1.041e-04	3.699e-01
HIRA	5.796892	4.053315e-08	1.224e-04	3.699e-01
SPINT4	5.809237	3.765500e-08	1.224e-04	3.699e-01
BNC1	-5.863049	2.726659e-08	1.224e-04	3.699e-01
TRPV1	5.692094	7.529438e-08	1.895e-04	4.771e-01
RPL23	5.516157	2.078961e-07	4.486e-04	6.886e-01
ATAD3C	5.485062	2.479943e-07	4.682e-04	6.886e-01
CLDN16	5.441644	3.167459e-07	5.021e-04	6.886e-01
PPP4C	5.433043	3.324059e-07	5.021e-04	6.886e-01
SNRPD1	-5.345258	5.417301e-07	7.438e-04	6.886e-01
AFP	5.283333	7.611261e-07	8.547e-04	6.886e-01
TUBB8B	5.274319	7.994997e-07	8.547e-04	6.886e-01
TRA2B	5.263323	8.488468e-07	8.547e-04	6.886e-01
TUBB8	5.293369	7.204990e-07	8.547e-04	6.886e-01
CELA3A	5.189181	1.267324e-06	9.658e-04	6.886e-01
SULT1A2	5.213602	1.111251e-06	9.658e-04	6.886e-01
UBE2H	-5.188551	1.271622e-06	9.658e-04	6.886e-01
H3-3A	5.214582	1.105395e-06	9.658e-04	6.886e-01
RPS14	5.187487	1.278905e-06	9.658e-04	6.886e-01
UBE2D1	-5.154457	1.526204e-06	1.098e-03	6.886e-01
ANOS	5.145556	1.600374e-06	1.099e-03	6.886e-01
UBE2B	5.105996	1.974339e-06	1.297e-03	6.886e-01
PGBD1	-5.096869	2.071900e-06	1.304e-03	6.886e-01
SF3B6	-5.016845	3.151613e-06	1.831e-03	6.886e-01

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
IQGAP3	-6.491090	5.113050e-10	7.723e-06	1.166e-01
HIRA	5.796892	4.053315e-08	1.224e-04	3.699e-01
SPINT4	5.809237	3.765500e-08	1.224e-04	3.699e-01
BNC1	-5.863049	2.726659e-08	1.224e-04	3.699e-01
PROM1	5.975173	1.379080e-08	1.041e-04	3.699e-01
TRPV1	5.692094	7.529438e-08	1.895e-04	4.771e-01
CEMIP	-4.964176	4.139594e-06	1.955e-03	6.886e-01
AFP	5.283333	7.611261e-07	8.547e-04	6.886e-01
SF3B6	-5.016845	3.151613e-06	1.831e-03	6.886e-01
TAS2R10	4.691656	1.626016e-05	3.411e-03	6.886e-01
PCLS	4.705640	1.518423e-05	3.354e-03	6.886e-01
BPIFB1	4.985994	3.698642e-06	1.908e-03	6.886e-01
SNAP25	-4.929729	4.940626e-06	1.987e-03	6.886e-01
NARS1	4.963700	4.149757e-06	1.955e-03	6.886e-01
SDS	4.754903	1.191839e-05	2.994e-03	6.886e-01
PGBD1	-5.096869	2.071900e-06	1.304e-03	6.886e-01
UBE2K	4.730274	1.345303e-05	3.209e-03	6.886e-01
CFAP20	-4.839052	7.827600e-06	2.627e-03	6.886e-01
CLDN16	5.441644	3.167459e-07	5.021e-04	6.886e-01
UBE2B	5.105996	1.974339e-06	1.297e-03	6.886e-01
ADAM2	4.990673	3.610160e-06	1.908e-03	6.886e-01
PPP4C	5.433043	3.324059e-07	5.021e-04	6.886e-01
PKD1L1	-4.800859	9.107662e-06	2.711e-03	6.886e-01
CELA3A	5.189181	1.267324e-06	9.658e-04	6.886e-01
ACTC1	-4.691812	1.624779e-05	3.411e-03	6.886e-01

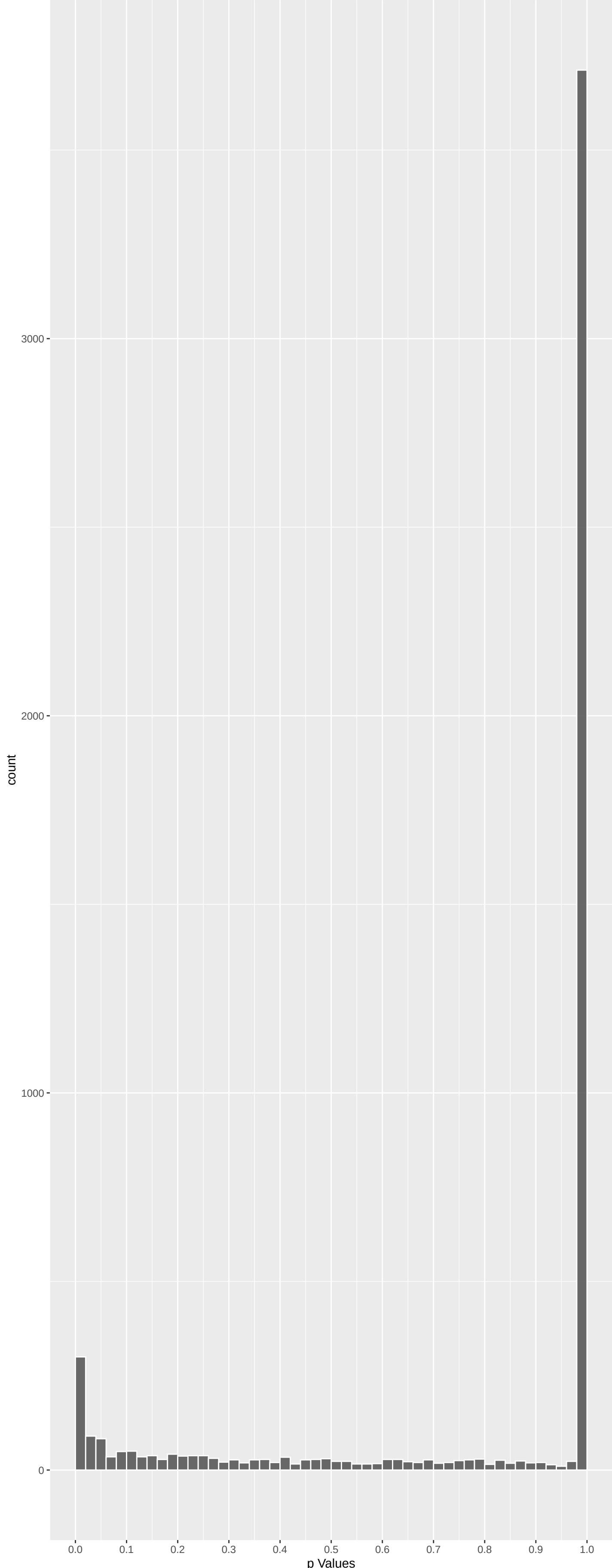
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
PROM1	5.975173	1.379080e-08	1.041e-04	3.699e-01
HIRA	5.796892	4.053315e-08	1.224e-04	3.699e-01
SPINT4	5.809237	3.765500e-08	1.224e-04	3.699e-01
TRPV1	5.692094	7.529438e-08	1.895e-04	4.771e-01
RPL23	5.516157	2.078961e-07	4.486e-04	6.886e-01
ATAD3C	5.485062	2.479943e-07	4.682e-04	6.886e-01
CLDN16	5.441644	3.167459e-07	5.021e-04	6.886e-01
PPP4C	5.433043	3.324059e-07	5.021e-04	6.886e-01
AFP	5.283333	7.611261e-07	8.547e-04	6.886e-01
TUBB8B	5.274319	7.994997e-07	8.547e-04	6.886e-01

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
IQGAP3	-6.491090	5.113050e-10	7.723e-06	1.166e-01
BNC1	-5.863049	2.726659e-08	1.224e-04	3.699e-01
SNRPD1	-5.345258	5.417301e-07	7.438e-04	6.886e-01
UBE2H	-5.188551	1.271622e-06	9.658e-04	6.886e-01
UBE2D1	-5.154457	1.526204e-06	1.098e-03	6.886e-01
PGBD1	-5.096869	2.071900e-06	1.304e-03	6.886e-01
SF3B6	-5.016845	3.151613e-06	1.831e-03	6.886e-01
UBE2I	-5.019001	3.116453e-06	1.831e-03	6.886e-01
CEMIP	-4.964176	4.139594e-06	1.955e-03	6.886e-01
SNAP25	-4.929729	4.940626e-06	1.987e-03	6.886e-01

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Steroid Metabolic Process (GO:0008202)	0.15944633	56	3.767e-05	2.022e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 CYP51A1:375 HSD17B3:409 CYP21A2:412 CYP27A1:499
'De Novo' AMP Biosynthetic Process (GO:0008202)	-0.15294345	4	2.895e-01	1.980e-01	PFAS:128 PAICS:5819 ATIC:5819 ADS51:5819 NA NA NA
'De Novo' Post-Translational Protein Fold (GO:0008202)	0.09781599	15	1.898e-01	1.980e-01	SDF2L1:887 DNAJB13:1097 HSPA1A:1742 GAK:1896 ERO1A:2443 ENTPD5:6847 PTGES3:6847
2-Oxoglutarate Metabolic Process (GO:0008202)	0.15259091	13	6.685e-02	1.980e-01	GOT2:166 TAT:169 OGDHL:185 GOT1:484 L2GHOD:1540 DHI:6847 PHYH:6847
3'-UTR-mediated mRNA Destabilization (GO:0008202)	-0.02404704	11	7.825e-01	1.947e-01	DND1:1427 TRIM71:5819 RC3H1:5819 ZFP36L1:5819 ZFP36L2:5819 MOV10:5819 ZFP36:5819
3'-UTR-mediated mRNA Stabilization (GO:0008202)	0.07242310	11	4.057e-01	1.980e-01	MAPK14:1303 HNRNPCL:1409 HNRNPFO:1754 DAZL:2202 ZFP36:6847 TARDBP:6847 YBX3:6847
3'-Phosphoadenosine 5'-Phosphosulfate Me (GO:0008202)	0.24757905	8	1.532e-02	1.980e-01	SULT1A2:97 SULT1B1:964 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 PAPSS2:6847 TPST2:6847
5S Class rRNA Transcription By RNA Polym (GO:000945)	-0.04055290	5	7.535e-01	1.980e-01	GTF3C2:5819 GTF3C3:5819 GTF3C4:5819 GTF3C5:5819 GTF3C6:5819 NA NA NA
7-Methylguanosine RNA Capping (GO:000945)	0.20867202	2	3.067e-01	1.980e-01	RNMT:569 CMTR2:6847 NA NA NA NA NA
7-Methylguanosine mRNA Capping (GO:000945)	0.20867202	2	3.067e-01	1.980e-01	RNMT:569 CMTR2:6847 NA NA NA NA NA
ADP Transport (GO:0015866)	-0.02679134	5	8.356e-01	1.980e-01	SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 SLC25A41:12017 NA NA
AMP Biosynthetic Process (GO:0006167)	-0.18086332	6	1.250e-01	1.980e-01	PFAS:128 APRT:855 PAICS:5819 ATIC:5819 ADS51:5819 NUDT2:5819 NA
ATF6-mediated Unfolded Protein Response (GO:0006167)	0.05763797	5	6.554e-01	1.980e-01	CREBZF:656 MBTPS1:6847 XBP1:6847 ATF6B:6847 DIT3:6847 NA NA
ATP Biosynthetic Process (GO:0006754)	0.02728418	13	7.335e-01	1.980e-01	TGFB1:780 ATP5F1C:1803 NUDT2:6847 VPS9D1:6847 ATP5PF:6847 ATP5PD:6847 ATP5PB:6847
ATP Metabolic Process (GO:0046034)	0.02856910	25	6.212e-01	1.980e-01	TGFB1:780 ENPP3:1053 BAD:1081 MYH6:1785 ATP5F1C:1803 PARG:2089 HSPA8:6847
ATP Synthesis Coupled Electron Transport (GO:0006754)	0.11562172	3	4.879e-01	1.980e-01	NDUF6:949 NDUF52:6847 NDUFY3:6847 NA NA NA NA
ATP Transport (GO:0015867)	-0.08235327	9	3.923e-01	1.980e-01	CALHM1:191 SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 CALHM3:5819 SLC17A9:5819
AV Node Cell To Bundle Of His Cell Commu (GO:0006754)	0.04574441	4	7.514e-01	1.980e-01	GJA5:2469 SCN5A:6847 CXADR:6847 GJC3:6847 NA NA NA
Arp2/3 Complex-Mediated Actin Nucleation (GO:0006754)	-0.09664430	10	2.900e-01	1.980e-01	WHAMM:727 ARPC5L:796 WASHC1:1166 ARPC1A:1319 ARPC5:5819 ARPC5:5819 ARPC2:5819
B Cell Activation (GO:0042113)	0.06407234	61	8.398e-02	1.980e-01	CD86:34 HDAC9:214 IFNK:316 IL4:679 CASP8:1036 CR2:1071 VCAM1:1120
B Cell Chemotaxis (GO:0035754)	0.05469964	5	6.617e-01	1.980e-01	CYP7B1:734 PIK3CD:6847 HSD3B7:6847 CXCL13:6847 GAS6:6847 NA NA
B Cell Differentiation (GO:0030183)	0.05402654	44	2.153e-01	1.980e-01	HDAC9:214 IFNK:316 IL4:679 CR2:1071 VCAM1:1120 TPTN2:1201 FLT3:3342
B Cell Homeostasis (GO:0001762)	0.03584862	7	7.426e-01	1.980e-01	TNFRSF13B:400 MEFC2:1121 GALT:6847 DCKA:1307 HNK1:6847 TNFAIP3:6847 DOK1:112293
B Cell Proliferation (GO:0042100)	0.07943161	15	2.870e-01	1.980e-01	IFNK:316 CR2:1071 MEFC2:1121 TPTN2:1201 IL6:6847 GALT:6847 IFNE:6847
B Cell Receptor Signaling Pathway (GO:0006754)	0.07446488	26	1.890e-01	1.980e-01	SH2B2:544 CTLA4:865 CD38:1042 MEFC2:1121 RFTN1:2025 PTPRC:2183 VAV3:2243
BMP Signaling Pathway (GO:0030505)	-0.04136061	42	3.542e-01	1.980e-01	LEFTY3:241 LEFTY4:198 GDF10:1012 FAM83C:1081 INHBE1:1119 SLC39A5:1205 SMAD5:1264
C-terminal Protein Amino Acid Modificat (GO:0006754)	-0.08813923	8	8.800e-01	1.980e-01	AGBL4:781 GLPD1:967 AGLB5:5819 AGTBP1:5819 FOLH1:5819 ATGL:15819 LCMT1:5819
C-terminal Protein Deglutamylation (GO:0006754)	-0.13997713	4	3.323e-01	1.980e-01	AGBL4:781 AGLB5:5819 AGTBP1:5819 FOLH1:5819 NA NA NA
C-terminal Protein Lysylation (GO:0006754)	-0.03478644	3	8.347e-01	1.980e-01	GLD1:967 ATG16L1:5819 ATG17:10875 NA NA NA
C4-dicarboxylate, Alpha-Beta T Cell Activati (GO:0006754)	0.11620399	10	2.033e-01	1.980e-01	SLCIA1:216 SLC25A12:618 SLC25A18:992 SLC13A2:1451 SLC25A13:6847 SLC13A5:6847 UCP2:6847
CD4-positive, Alpha-Beta T Cell Activity (GO:0006754)	-0.13020632	5	3.133e-01	1.980e-01	KNK7:147 STOM1:242 SNTF1:5819 TNFSF4:5819 RSDA2:5819 HMGCS1:5819 NA NA NA
CD40 Signaling Pathway (GO:0023035)	0.03632796	7	7.393e-01	1.980e-01	CD86:34 CD40LG:6847 ITGB1:6847 PHB2:6847 TNIP2:6847 ITGA5:6847 RNF31:6847
CENP-A Containing Chromatin Assembly (GO:0006754)	0.04377507	5	7.371e-01	1.980e-01	NASP:1553 M158A:6847 CENPL:6847 HUJRP:6847 NA NA NA NA
COP1 Coating Of Golgi Vesicle (GO:0034824)	-0.05684820	2	7.807e-01	1.980e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA NA
COP1-coated Vesicle Budding (GO:0005860)	-0.05684820	2	7.807e-01	1.980e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA NA
COP2 Coating Of Golgi Vesicle (GO:0034824)	-0.03438544	19	6.561e-01	1.980e-01	KLHL12:475 TRAPPC12:1193 TRAPPC3:5819 TRAPPC2:5819 TRAPPC1:5819 TRAPPC5:5819
COP2-coated Vesicle Budding (GO:0005860)	-0.03438544	19	6.561e-01	1.980e-01	KLHL12:475 TRAPPC12:1193 TRAPPC3:5819 TRAPPC2:5819 TRAPPC1:5819 TRAPPC5:5819
CRD-mediated mRNA Stabilization (GO:0070067)	-0.19756073	5	1.261e-01	1.980e-01	IGF2BP1:496 IYB1:1195 DHX9:5819 PAPBCL:5819 HNRNPDP:5819 NA NA NA
Cdc42 Protein Signal Transduction (GO:0006754)	-0.04055702	6	7.308e-01	1.980e-01	SHTN1:5819 WAS:5819 RHOV:5819 NTN1:5819 RHOBTB1:5819 RHOBTB2:5819 NA NA
DNA Alkylation (GO:0006305)	0.09262387	15	2.144e-01	1.980e-01	DNMT3B:31 GATAD2A:146 BAZ2A:343 HELLS:1649 KMT2E:6847 MTRR:6847 ATRX:6847

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON	-0.29109711	27	1.661e-07	1.075e-03	ADAM15:45 BPXIP1:80 MTX1:85 RUSC1:89 RORC:177 FOPD5:188 THBS3:190
NIKOLSKY_BREAST_CANCER_Q2Q2_Q24_AMPLICON	-0.14502570	100	5.637e-07	1.825e-03	COL20A1:300 HELL23:88 MTG2:97 CDH2B:104 FAM217B:130 LAMA5:149 SLC0A41:210
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.13697150	92	5.790e-06	1.250e-02	SULT1A2:97 AKR1D11:177 MGST1:161 CYP27B1:176 COMT:179 CYP11A1:201 GSTK1:235
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.23916671	26	2.444e-05	3.957e-02	HSD17B2:14 HSD11B1:107 AKR1D1:117 COMT:179 CYP11A1:201 HSD17B3:409 CYP21A2:412
REACTOME_BIOLOGICAL_OXIDATIONS	0.09961207	123	1.503e-04	1.947e-01	ALDH1A1:73 SULT1A2:97 HPGDS:136 MGST1:161 CYP27B1:176 COMT:179 CYP11A1:201
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.19383671	28	3.868e-04	2.784e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 SERPINA6:585 CYP17A1:488
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS	0.10671382	95	3.319e-04	2.784e-01	SLC36A1:71 SLC36A2:135 SLC12A3:193 SLC1A1:216 SLC6A6:251 SLC7A6:272 SLC20A1:183
REACTOME_METABOLISM_OF_LIPIDS	0.04463701	576	2.908e-04	2.784e-01	HSD17B2:14 SLC01B1:127 PGTG1:96 HSD11B1:107 AKR1D1:117 HPGDS:136 RORA:170
CAIRO_HEPATOBLASTOMA_DN	0.07405818	196	3.441e-04	2.784e-01	TMEM131L:35 AKR1D1:117 SLC46A3:125 CA2:152 SLC7A5:177 OGDHL:185 AGXT:186
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.07628784	180	4.324e-04	2.800e-01	IFNAR1:58 IL2R2:66 TNF:87 TNFSF10:103 TNFSF15:112 IL1RAP:304 IFNK:316
NIKOLSKY_BREAST_CANCER_Q2Q3_Q24_AMPLICON	-0.09228699	111	7.998e-04	3.176e-01	PLEC:24 FAM83H:202 GPR20:254 TRAPP9:286 OLAH:293 SCRIB:348 EEF1D:409
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.04346604	5	8.339e-04	3.176e-01	CA2:152 CA4:253 AQP1:762 SLC1A1:1060 RHAG:2146 NA NA
WP_Q2Q12_COPY_NUMBER_VARIATION_SYNDROME	0.06778672	219	5.733e-04	3.176e-01	SLC22A16:3 SLC36A1:71 SLC01B1:82 SLC10A6:83 SLC36A2:135 SLC12A3:193 SLC10A2:198
WP_STEROID_HORMONE_PRECURSOR_BIOSYNTHESI	0.17941272	30	6.742e-04	3.176e-01	CNNM4:163 ARID5A:424 CNNM3:508 KANSL3:580 COL2A1:605 DUSP2:677 FAHD2B:759
COP1-coated Vesicle Budding (GO:0005860)	-0.05684820	2	7.807e-01	1.980e-01	AKR1D1:117 CYP11A1:201 CYP21A2:412 CYP17A1:883 HSD3B2:1180 SRD5A2:1125 SRD5A1:6847
KEGG_ALANINE_ASPARTATE_AND_GLUTAMATE_MET	0.20027634	24	6.854e-04	3.176e-01	GOT2:166 AGXT:186 CAD:245 IL4I1:373 GLS2:427 GOT1:484 ASL:589
KEGG_CYSSTEINE_AND_METHIONINE_METABOLISM	0.21228245	21	7.602e-04	3.176e-01	DNMT3B:31 DNMT1:113 GOT2:166 TAT:169 IL4I1:373 LDHC:408 GOT1:484
NIKOLSKY_BREAST_CANCER_IQ24_AMPLICON	-0.14706730	40	1.297e-03	3.989e-01	FTSJ3:58 HOXB13:67 LRRC59:71 ITGBA:100 ENGASE1:19 EVPL:121 SNF:189
BENPORATH_HS_WITH_H3K27ME3	-0.03573796	730	1.117e-03	3.810e-01	HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 CYP17A1:883 HSD3B2:1180 CYP19A1:2015
REACTOME_GLUCCOCHONDIAC_BIOSYNTHESIS	0.36870547	6	1.762e-03	4.226e-01	CEMP10:0 BNC1:14 PLEC:24 SUSD4:34 ADAMTSL3:50 HOXB13:67 DACH2:75
WP_ALLOGRAFT_REJECTION	0.11271893	65	1.691e-03	4.226e-01	HSD11B1:107 CYP21A2:412 SERPINA6:585 CYP17A1:883 HSD3B2:1180 HSD11B2:6847 NA
WP_PROXIMAL_TUBULE_TRANSPORT	0.13658419	44	1.732e-03	4.226e-01	CD86:34 TNF:87 HLA-DQB1:137 FOXO3:324 ABCB1:397 IL1A:608 IL4:679
KEGG_SULFUR_METABOLISM	0.30192728	9	1.710e-03	4.226e-01	ATP6V0A4:43 SLC36A2:135 CA2:152 SLC1A1:216 CA4:253 ABCB1:397 SLC5A8:692
KEGG_TYPE_1_DIABETES_MELLITUS	0.18085786	25	1.753e-03	4.226e-01	SULT1A2:97 SULT1A4:370 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 CHST13:2118 SUOX:2221
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	-0.04988831	333	1.866e-03	4.316e-01	CD86:34 TNF:87 HLA-DQB1:137 FOXO3:324 ABCB1:397 IL1A:608 IL4:679
REACTOME_ARACHIDONIC_ACID_METABOLISM	0.14099083	40	2.044e-03	4.564e-01	CHRN2B:26 SHE4D CAMK2B:46 HTR1D:55 KCHN2:81 YBX2:87 ZBTB16:98
REACTOME_CYTOCHROME_P450_ARRANGED_BY_SUB	0.14731339	35	2.573e-03	5.375e-01	PTGR1:96 HPGDS:136 GGT1:470 GPX4:500 CYP1A1:528 GGT5:642 GPX1:786
KEGG_GRAFT_VERSUS_HOST_DISEASE	0.21133524	17	2.559e-03	5.375e-01	CYP27B1:176 CYP11A1:201 CYP51A1:375 CYP21A2:412 CYP27A1:499 CYP1A1:528 CYP7B1:734
PID_NFAT_TFPATHWAY	0.14661319	35	2.697e-03	5.458e-01	CD86:34 TNF:87 HLA-DQB1:137 FOXO3:324 ABCB1:397 IL1A:608 IL4:679
REACTOME_MEIOTIC_RECOMBINATION	-0.15454536	31	2.912e-03	5.714e-01	TNF:87 FOXO3:324 HGKA:353 EGR4:646 IL4:679 CTLA4:865 NFATC3:915
REACTOME_TNFS_BIND THEIR PHYSIOLOGICAL_R	-0.17720343	23	3.271e-03	6.186e-01	MRE11:406 ATM:434 DGA2:626 MHL1:818 H3-4:831 HTRC9:1058 TOP3A:1215
PYEON_HVP_POSITIVE_TUMORS_UP	0.09915087	73	3.436e-03	6.186e-01	TNFSF15:112 TNFRSF13B:400 TNFSF14:613 TNFSF78:160 TNFRSF9:836 FASLG:1216 LTA:1294
KEGG_ALLOGRAFT_REJECTION	-0.18454841	21	3.439e-03	6.186e-01	TASOR:41 TAT:169 UHRF1:155 TRIM56:411 DTL:423 GLS2:427 NR1D2:478
LASTOWSKA_NEUROBLASTOMA_COPY_NUMBER_UP	-0.07130891	138	3.908e-03	6.214e-01	CD86:34 TNF:87 HLA-DQB1:137 IL4:679 CD80:787 IFNG:1016 IL2B:1171
BIOCARTA_HBX_PATHWAY	-0.36844491	5	4.327e-03	6.214e-01	MINK1:51 FTSJ3:58 LRRC59:71 NEURL4:200 CERN1:265 PAHB:270 DPH1:276
IVANOVA_HEMATOPOIESIS_MATURE_CELL	0.05934998	203	3.673e-03	6.214e-01	SHC1:353 CREB1:565 LAMTOR5:675 SOS1:995 PTX2B:5819 NA NA
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	-0.16105691	26	4.487e-03	6.214e-01	ST3GALS4:81 IL33:127 GONT1:129 TSPAN1:67 TM4SF4:195 COL1A1:209 CPD:223
					TP53BP2:38 ATM:434 CASP2:465 AIFM2:528 TP73:610 PERP:828 PIDD1:1085

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Dermatitis	0.10566782	138	1.950e-05	6.349e-02	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151 PLAUR:164
Leprosy	0.13214844	88	1.908e-05	6.349e-02	TNF:87 TNFSF15:112 HLA-DQB1:137 FOXO3:324 IL13RAP:416 CD163:524 IL1A:608
Malignant neoplasm of colon stage IV	0.23651638	29	1.347e-05	6.349e-02	PROM1:22 TNFSF10:103 TPX2:286 SMPD1:606 FBLN1:633 VEGFA:756 TNFSF9:836
Hepatitis C	0.05052428	498	3.112e-05	7.599e-02	AFP:1 CLU:17 GTGLC3:29 STAT2:38 IFNAR1:58 IL2R2:66 TNF:87
Severe Dengue	0.19005259	38	4.749e-05	9.275e-02	RBFOX1:36 TNF:87 TRT:238 VEGFA:756 CTLA4:865 CXCL8:1051 ICAM1:1079
Cystic Fibrosis	0.06059623	373	6.930e-05	9.668e-02	AFP:1 BPIFB1:4 DNAMT3B:31 CELA3B:42 TRPV5:55 TNF:87 HPGDS:136
Systemic Scleroderma	0.06286578	347	6.705e-05	9.668e-02	PRSS55:2 IFNAR1:58 ENG:64 TNF:87 TNFSF10:103 DNMT1:113 IL33:127
Psoriasis	0.05100856	522	8.434e-05	1.030e-01	IL1RL2:8 STAT2:38 TNF:87 TNFSF10:103 TNFSF10:103 TNFSF15:112 IL33:127
Balkan Nephropathy	0.35348346	10	1.094e-04	1.189e-01	TNF:87 GSTK1:235 ABCB1:397 ALB:530 SLC6A1:599 IL1A:608 CTR:672R1:819
Goldenhar Syndrome	-0.36617357	9	1.427e-04	1.393e-01	TSC1:59 FRAS1:279 TBC1D20:31 FGF8:316 GRIP1:538 CHD7:768 SALL1:1306
Hepatitis	0.07592066	201	2.231e-04	1.676e-01	AFP:1 PROM1:22 SLC01B1:82 TNF:87 TNFSF10:103 AKR1D1:117 IL33:127
IGA Glomerulonephritis	0.08276435	169	2.192e-04	1.676e-01	CD86:34 TNF:87 ITGAX:104 MAN2B1:108 HORMAD2:118 FCGRT:149 MEV3:362
Inflammatory dermatitis	0.10099228	115	1.913e-04	1.676e-01	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151 TRPA1:183
Renal salt wasting	0.28288815	14	2.484e-04	1.733e-01	CYP11A1:201 CYP21A2:412 SCN1A:465 CLCNKA:664 SCN1G:1157 HSD3B2:1180 CLCNKB:1477
Kidney Failure, Chronic	0.05924546	310	3.741e-04	1.857e-01	TRPV1:13 CLU:17 CHD7:77 TNF:87 HSD11B1:107 MNT:115 PDXP:143
Acquired aplastic anemia	0.17905557	34	3.056e-04	1.857e-01	TNF:87 HPGDS:136 GSTK1:235 FOXO3:324 HNFI:559 SLC6A1:599 IL1A:608
Anemia	0.05613830	344	3.926e-04	1.857e-01	DNMT3B